

16. **Ans: C**

$$\frac{1}{2} m(c_{\text{rms}})^2 = \frac{3}{2} kT$$

$$m(c_{\text{rms}})^2 \propto T$$

$$\frac{m_{\text{He}} (c_{\text{rms}})^2_{\text{He}}}{m_{\text{Ne}} (c_{\text{rms}})^2_{\text{Ne}}} = 1 \quad (\text{thermal equilibrium})$$

$$\frac{(c_{\text{rms}})^2_{\text{He}}}{(c_{\text{rms}})^2_{\text{Ne}}} = \frac{m_{\text{Ne}}}{m_{\text{He}}}$$

$$\frac{c_{\text{rmsHe}}}{c_{\text{rmsNe}}} = \sqrt{\frac{m_{\text{Ne}}}{m_{\text{He}}}}$$

$$= \sqrt{2.5}$$

$$= 1.6$$